

Problem set 5:

Q 1) To prove: $\dim(R)$
a)

dimension of vector space R
over field R is 1.

Proof: Basis: i) spans set
ii) is linearly independent.

i). As $\forall x \in R$ (vector space)
 $x \in R$ (field)
 $x = x$
 $\therefore x = 1 \cdot x$.

$\therefore 1$ spans vector space R

$\therefore \forall y \in R$

$$y = 1 \cdot y \Rightarrow y = 1 \cdot c,$$

$$c \in R$$

$$c = y.$$

$$\therefore \text{span } \{1\} = R.$$

(1)

ii) for linearly independent

$$V = \sum v_1, v_2, \dots, v_n$$

$$c_1 v_1 + c_2 v_2 + \dots + c_n v_n = 0$$

~~iff~~ iff $c_1, c_2, \dots$
are all
zero

for $c(1) = 0$.

$$\boxed{c = 0}$$

$\therefore$ $\{1\}$ is linearly independent

$\therefore$ from (1) & (2),

$\{1\}$ is the basis of R over K .

b) Dimension of vector space R over
To prove: field $\mathbb{Q}$ is infinite.

i.e., There is no finite basis
of R over $\mathbb{Q}$.

Proof:

for prime no. s (distinct),

$$p_1, p_2, \dots, p_n,$$

Prone: ~~is~~ $\log^{p_1}, \log^{p_2}, \dots$

~~is~~ are linearly independent over $\mathbb{Q}$.

Proof: for $q_i \in \mathbb{Q}, i \in [1, n]$,

$$q_1 \log^{p_1} + q_2 \log^{p_2} + \dots + q_n \log^{p_n} = 0,$$

$$q \in \mathbb{Q}.$$

Now, ~~divide~~ multiply both sides with ~~the~~.

let ~~q~~ a_r be common denominator of

$$q_i, i \in [1, n]$$

Multiplying with $a_r, a_r \in \mathbb{Z}$

on both sides

∴ It is now equal to

$$a_1 \log p_1 + a_2 \log p_2 + \dots + a_n \log p_n$$

$$= 0$$

$$a_i \in \mathbb{Z}.$$

$$\text{as } q_i = \frac{a_i}{a_i}.$$

$$\boxed{\arg q_i = a_i}$$

$$\text{Now } y \log^n = \log^{ny}$$

$$\therefore \log^{p_1^{a_1}} + \log^{p_2^{a_2}} + \dots + \log^{p_n^{a_n}} = 0.$$

$$\text{As } \log^a + \log^b = \log^{ab},$$

$$\text{and } \log^1 = 0,$$

⇒

we get,

$$\log^{(p_1)^{a_1} (p_2)^{a_2} \dots (p_n)^{a_n}} = 0$$

$$= \log^1$$

$$\therefore (p_1)^{a_1} (p_2)^{a_2} \dots (p_n)^{a_n} = 1$$

But, p_i are primes, $-(p_i)^{a_i}$ prime,

Which means, product of

~~primes~~ + coprimes
iff all of them
are 1.

$$(p_i)^{a_i} = 1$$

$$\therefore a_i = 0 \text{ as } p_i \neq 1$$

1 & 2 primes?

$$\therefore a_i = 0 \quad \forall i \in \{1, n\}.$$

$$\therefore q_1 = q_2 = \dots = q_n = 0.$$

$$A \subseteq \{ \log p_1, \log p_2, \dots, \log p_n \}$$

is linearly independent over $\mathbb{Q}$.

but no. of primes is infinite.

$$\text{Hence, } |A| = \infty.$$

$$\therefore R \text{ has, } \infty$$

so no. of such sets
 (finite dependent sets) of R
 over $\mathbb{Q}$ is
 arbitrarily large.

$$\therefore \boxed{\lim_{\theta} \frac{f}{g} = \infty.}$$

28. Given: V is an n -dimensional vector space over F .

β and β' are two ordered bases of V .

To prove: If $[x]_{\beta} = [x]_{\beta'} \quad \forall x \in V$,

$$\forall x \in V,$$

prove that $\beta = \beta'$.

$$[x]_{\beta} = (c_1, c_2, \dots, c_n).$$

$$\therefore [x]_{\beta} = c_1 v_1 + c_2 v_2 + \dots + c_n v_n$$

$$[x]_{\beta'} = c_1 w_1 + c_2 w_2 + \dots + c_n w_n.$$

$$\beta = \{v_1, v_2, \dots, v_n\}$$

$$\beta' = \{w_1, w_2, \dots, w_n\}.$$

$$v_i = \cancel{0(w_1)} + \cancel{0(w_2)} + \dots$$

Let $\alpha = v_i \quad v_i \in \beta$

$$\therefore [\alpha]_{\beta} = [v_i]_{\beta} = e_i$$

where e_i

$$= (0, 0, \dots, 1, 0, \dots, 0)$$

$\swarrow$ i th position,

Now, $[\alpha]_{\beta} = [\alpha]_{\beta'}$

$$\therefore [v_i]_{\beta'} = e_i$$

$$\therefore v_i = 0w_1 + 0w_2 + \dots + 1(w_i) + \dots + 0(w_n)$$

$v_i = w_i$

so, $\forall v_i \in \beta$

$$v_i = w_i, \quad w_i \in \beta'$$

$$|\beta| = |\beta'| = \text{~~dim~~} = n$$

Hence

$$\boxed{\beta = \beta'}$$

as all

bases

have

same

~~dimension~~ dimension

37. Given: V is an n -dimensional vector space over field F ,

$$\alpha_1, \alpha_2, \dots, \alpha_n \in V$$

are linearly independent.

To prove: $\{\alpha_1, \alpha_2, \dots, \alpha_n\}$ are a basis of V .

Proof: If W is a proper subspace of V ,

$$\dim(W) < \dim(V)$$

Let $W = \text{span}\{\alpha_1, \alpha_2, \dots, \alpha_n\}$.

as $\alpha_1, \alpha_2, \dots, \alpha_n$ are linearly independent,

$$\boxed{\dim(W) = n.} \quad \textcircled{1}$$

Now, as $\alpha_1, \alpha_2, \dots, \alpha_n \in V$,

$$W \subseteq V.$$

If $W \neq V$, then $\boxed{\dim W < \dim V}$.

But, $\boxed{\dim(W) = n}$ from (i)

$$\boxed{\dim(V) = n} - \text{given.}$$

$$\therefore \dim W = \dim V.$$

$\therefore$ Proof by contradiction,

$$\boxed{W \not\subseteq V.}$$

$\therefore$ as $\{ \alpha_1, \alpha_2, \dots, \alpha_n \}$

span V and are linearly independent,

they form the basis of V .

4). Given: Let V be an n -dimensional vector space over field F .

Let $\alpha_1, \alpha_2, \dots, \alpha_n \in V$

span V . ~~Show~~

To prove: $\{\alpha_1, \dots, \alpha_n\}$ is a basis of V .

Proof: Proof by Contradiction

Let $\{\alpha_1, \dots, \alpha_n\}$ be a linearly dependent set.

$\therefore$ ~~as remaining~~

$$\alpha_i = c_1 \alpha_1 + c_2 \alpha_2 + \dots + c_n \alpha_n$$

for some i .

$\therefore \{\alpha_1, \dots, \alpha_n\} / \alpha_i$ is also a span of V

But then, number of vectors in span V is $n-1$.

But a spanning set
MUST have $\geq n$ vectors,

$\therefore$ The assumption is false.

$\{2, \dots, n\}$ is
linearly independent.

As it's linearly independent
and spans V , it forms
the basis of V .

5). P_2 is the vector space of all
polynomials with degree ≤ 2
over field R .

$$B = \{1, x, x^2\}$$

$$B' = \{1+x, 2+3x, 5+2x+x^2\}$$

to ~~find~~ ^{find}: B and B' are ordered
bases of P^2 .

to ~~find~~ ^{Find} matrix P such
that

$$[p(x)]_B = P [p(x)]_{B'}$$

$$\forall p(x) \in P_2.$$

Solⁿ:

$$\beta' = (1+n, 2+3n, 5+n+n^2)$$

$\therefore \beta'$ in terms of B

$$1+n = (1, 1, 0)$$

$$2+3n = (2, 3, 0)$$

$$5+n+n^2 = (1, 1, 5)$$

$$\therefore A' = \begin{bmatrix} 1 & 2 & 5 \\ 1 & 3 & 1 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\text{Now, } \det(A') = 1(3(1) - 1(0))$$

$$+ 1((0)(5) - 2(1))$$

$$+ 0$$

$$= 3 - 0 - 2 = \boxed{1}$$

$$1 \neq 0$$

$\therefore \beta'$ is linearly independent.

As it also spans P ,

β' is a basis of P .

Now, Let $P = B' \text{ matrix } A'$

$$= \begin{bmatrix} 1 & 2 & 5 \\ 1 & 3 & 1 \\ 0 & 0 & 1 \end{bmatrix}$$

for (a, b, c)

$$\text{Now } p(n) \beta^{\#} = \begin{bmatrix} a \\ b \\ c \end{bmatrix}$$

$$p(n) = a(1+n) + b(2+3n) + c(5+n+n^2)$$

$\therefore$ relative to β , it becomes

$$a \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} + b \begin{bmatrix} 2 \\ 3 \\ 0 \end{bmatrix} + c \begin{bmatrix} 5 \\ 1 \\ 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 2 & 5 \\ 1 & 3 & 1 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \end{bmatrix} = P C p(n) \beta'$$

$$\therefore [p(n)]_{\beta} = P[p(n)]_{\beta'}$$

6). Given : $\mathbb{R}^4$ is defined by

$$\alpha_1 = (-1, 0, 1, 2)$$

$$\alpha_2 = (3, 4, -2, 5)$$

$$\alpha_3 = (1, 4, 0, 9)$$

To find : Homogeneous system of linear equations
for which solution space =
Subspace of $\mathbb{R}^4$ spanned
by $\alpha_1, \alpha_2, \alpha_3$.

$\therefore$ Solⁿ set = span $\{\alpha_1, \alpha_2, \alpha_3\}$.

Solⁿ :

for matrix $A = \begin{bmatrix} -\alpha_1 & - \\ -\alpha_2 & - \\ -\alpha_3 & - \end{bmatrix}$

$$A = \begin{bmatrix} -1 & 0 & 1 & 2 \\ 3 & 4 & -2 & 5 \\ 1 & 4 & 0 & 9 \end{bmatrix}$$

$$R_1 \leftarrow -R_1$$

$$R_2 \leftarrow R_2 - A_{11}R_1, A_{21}R_1$$

$$R_1 \leftarrow -R_1$$

$$R_2 \leftarrow R_2 + 3R_1$$

$$R_3 \leftarrow R_3 + R_1$$

$$A_2 = \begin{bmatrix} 1 & 0 & -1 & -2 \\ 0 & 4 & 1 & 1 \\ 0 & 4 & 1 & 1 \end{bmatrix}$$

$$R_2 \leftarrow R_2 / 4$$

$$R_3 \leftarrow R_3 - R_2$$

$$A_2 = \begin{bmatrix} 1 & 0 & -1 & -2 \\ 0 & 1 & 1/4 & 1/4 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

rank = 2

$\therefore$ dimension of span = 2

$\therefore$ from rref:

$$x_1 - x_3 - 2x_4 = 0$$

$$x_2 + \frac{x_3}{4} + \frac{11x_4}{4} = 0$$

$\therefore$ final system of homogeneous linear equations:

$$x_1 - x_3 - 2x_4 = 0$$

$$4x_2 + x_3 + 11x_4 = 0$$

7) Given: R is an $m \times n$ RREF matrix over field F .

Non zero rows: $p_1, p_2, \dots, p_r$.

W is its row space.

$\beta = (b_1, b_2, \dots, b_n)$ is in W .

To find: i). coefficients $c_1, \dots, c_r$
 in terms of $b_1, \dots, b_n$
 When $\beta = c_1 p_1 + \dots + c_r p_r$

ii). Let $s \in \{1, r\}$

To prove: $\exists! \alpha_s \in W$

such that b_s is a coordinate of $\alpha_s = 1$ and

b_i coordinates of α_s are 0.

($i \in \{1, r\} \setminus \{s\}$)

iii). Is $\alpha_s = p_s$. Prove it.

SD

i). For every row p_i , there's a leading column i such that $A_{p_i i} \neq 0 \quad \forall p_i \neq p_j$

$\therefore \beta = \sum c_i p_i$ However, for all $p_i \neq p_j$
 $A_{p_i i} \neq 0$

$$\therefore b_{ki} = c_i$$

$$c_i = b_{ki}$$

~~any~~

2.2) a set such that

its' k th coordinate = 1
 other point coordinates = 0.

To make point $k = 1$,

set: $c_k = 1$.

else, $c_i = 0$ ($i \neq k$)

$\therefore c_k = 1, c_i = 0$ ($i \neq k$)

$\therefore a_s = p_s$.

Which makes it a unique vector.

iii) $q_s = p_s$?

Yes. if $c_i = 0$ and $c_s = 1$
 $i \neq s$

$$a_s = 0 p_1 + 0 p_2 + \dots + 1 p_s + \dots + 0 p_r$$

$$\boxed{a_s = p_s}$$

8) Given: W is a subspace of F^n
 $\dim W \leq n$.

$\beta = (b_1, \dots, b_n)$ is in W .

$T = \{k_1, \dots, k_r\} \quad \forall t \in T,$
 $\exists \beta \neq 0 \in W$

$$\beta = (0, \dots, 0, b_t, \dots, b_n)$$

To prove:

i) $k_1 < k_2 < \dots < k_r$

ii) $k_i \neq k_j$

$$\beta_1 = (0, \dots, 0, b_{k_1}, \dots, b_r)$$

$$\beta_2 = (0, \dots, 0, b_{k_2}, \dots, b_r)$$

are linearly independent.

iii). for each k_s , $\exists$ one vector x_s in W . both coordinates of $x_s \neq 1$.

both coordinates of $x_s \neq 0$
 $i \neq s$.

Proof:

i). As sets are unordered, order doesn't matter.

Hence, we can say

$k_1 < k_2 < \dots < k_n$
without loss of generality

ii). Let $k_i < k_j$

Proof by contradiction:

β_1 and β_2 are dependent.

$\therefore$ for $a \neq 0$ and $b \neq 0$.

$$a\beta_1 + b\beta_2 = 0$$

for coordinate k_i ,

β_2 has $b_{ki} = 0$ as $k_i < k_j$

$$\therefore a b_{ki} = 0$$

but $b_{ki} \neq 0 \rightarrow$ definition

$$\therefore \boxed{a = 0}$$

$$\text{So, } 0(\beta_1) + b(\beta_2) = 0$$

$$b\beta_2 = 0$$

$$\beta_2 \neq 0$$

$$\therefore \boxed{b = 0}$$

$\therefore$ by contradiction,

β_1 and β_2 are linearly independent.

iii.

$$\beta_s = (0, \dots, 0, b_s, \dots)$$

$$b_s \neq 0.$$

$$\gamma_s = \frac{\beta_s}{b_s}$$

$\therefore$ k_s^{th} coordinate = 1.

via gaussian elimination,

~~β_s~~

$$\gamma_s = (0, \dots, 0, b_s, 0, \dots)$$

Now, let γ_s and γ_s'
be two such
vectors,

for all k_i in γ_s and γ_s'

$$\gamma_s[k_i] = \gamma_s'[k_i]$$

$$\boxed{\gamma_s = \gamma_s'}$$

$$\therefore \delta = \gamma_s' - \gamma_s$$

but δ has 1st non-zero
coordinate on γ_s / $\delta \neq 0$

But by definition,

any non zero vector in W must have its first non zero coordinate in T .

$$\therefore \delta = 0$$

$$\nexists \delta \text{ s.t. } \gamma_s = \delta_s'$$

9). To find: which are linear transformations

i). $T(n_1, n_2) = (n_1, (1+n_2))$

ii). $T(n_1, n_2) = (n_1^2, n_2)$

iii). $T(n_1, n_2) = (0, n_1 - n_2)$

Linear transformations have

$$T(0, 0) = (0, 0)$$

$$T(u+v) = T(u) + T(v)$$

$$T(cu) = cT(u)$$

a). $T(0, 0) = T(0, 1)$

But $T(0, 0) = (0, 0)$
as ~~0 is not 1~~

Hence, NOT a linear transformation

$\begin{pmatrix} 0 & 0 \\ 1 & 1 \end{pmatrix}$

$$u = (1, 0)$$

$$v = (1, 0)$$

$$T(u+v) = T(2, 0) = (4, 0)$$

$$T(u) + T(v) = (1, 0) + (1, 0) \\ = \boxed{(2, 0)}$$

Not a linear transformation

iii). $T(n_1, n_2) = (0, n_1 - n_2)$

$$T(0, 0) = (0, 0)$$

$$T(n_1 + a, n_2 + b) = (0, n_1 + a - n_2 - b)$$

$$\Rightarrow T(n_1, n_2) + T(a, b) \\ = (0, n_1 - n_2) + (0, a - b)$$

$$= (0, n_1 + a - n_2 - b)$$

So, it's a linear transformation.